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Brain MRI Segmentation

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Development of Advanced Computer Science Applications

Abstract

1. Introduction

2. Data Exploration

2.1 Dataset Introduction

The dataset source comes from "*Brain MRI segmentation*" by Mateusz Buda [1]. The dataset in turn is the LGG Segmentation Dataset, which focuses on the segmentation of abnormal regions in brain MRI images from patients diagnosed with lower-grade glioma (LGG). This dataset was built from medical imaging data obtained from The Cancer Imaging Archive (TCIA) and is linked to cases from The Cancer Genome Atlas (TCGA) lower-grade glioma collection. In addition to imaging data, the dataset also includes clinical and genomic information for each patient through the file data.csv.

This dataset is designed for the study of brain tumor abnormalities visible in FLAIR MRI sequences, and it is commonly used in medical image segmentation and radiogenomic analysis. The segmentation masks were created manually and serve as the ground truth for supervised learning tasks.

The dataset is especially relevant because it combines two important sources of information:

1. Medical imaging data, which supports the segmentation task.
2. Patient-level genomic and clinical data, which can be used for complementary analysis and future radiogenomic studies.

2.2 Structure of the Dataset

The dataset is organized into **110 folders**, each corresponding to one patient. Every folder contains multiple MRI slices and their corresponding segmentation masks. The files follow a naming convention that allows each MRI image to be matched directly with its mask.

The general naming format is:

TCGA_<institution-code>_<patient-id>_<slice-number>.tif

and the corresponding mask is:

TCGA_<institution-code>_<patient-id>_<slice-number>_mask.tif

This means that for each MRI slice there is an associated mask file with the same identifier and the suffix _mask.

2.3 File Types and Format

The images are stored in .tif format and each image has 3 channels. These channels correspond to different MRI sequences:

- pre-contrast
- FLAIR
- post-contrast

For some patients, one of these sequences is missing. In those cases, the missing sequence was replaced with the FLAIR sequence so that all images maintain a consistent 3-channel structure.

The segmentation masks are also stored in .tif format, but they are single-channel binary images. These masks indicate whether each pixel belongs to the abnormal region visible in the FLAIR sequence. In the explored dataset, the mask values are represented as 0 for background and 255 for abnormal region, which can be converted into 0 and 1 for training purposes.

2.4 Number of Patients and Samples (instances)

The dataset contains information for 110 patients. From the exploratory analysis, the dataset includes:

- 3929 MRI images
- 3929 segmentation masks

All MRI images have shape:

- $256 \times 256 \times 3$

All masks have shape:

- 256×256

Thus, the dataset is composed of 2D MRI slices rather than full 3D volumes, and each slice is paired with its corresponding segmentation mask.

2.5 Additional Information

Besides the image and mask files, the dataset includes a file called data.csv, which contains patient-level metadata. This file provides information such as:

- **Genomic subtype clusters:** molecular groupings of tumors based on genomic data, used to identify biologically similar glioma subtypes.
- **Histological type:** tissue-based classification describing the tumor according to its microscopic characteristics.
- **Tumor grade:** indicates the severity or aggressiveness of the tumor.
- **Tumor location:** anatomical region of the brain where the tumor is found.
- **Gender**: biological sex of the patient.
- **Age:** patient age at the time of diagnosis or pathological evaluation.

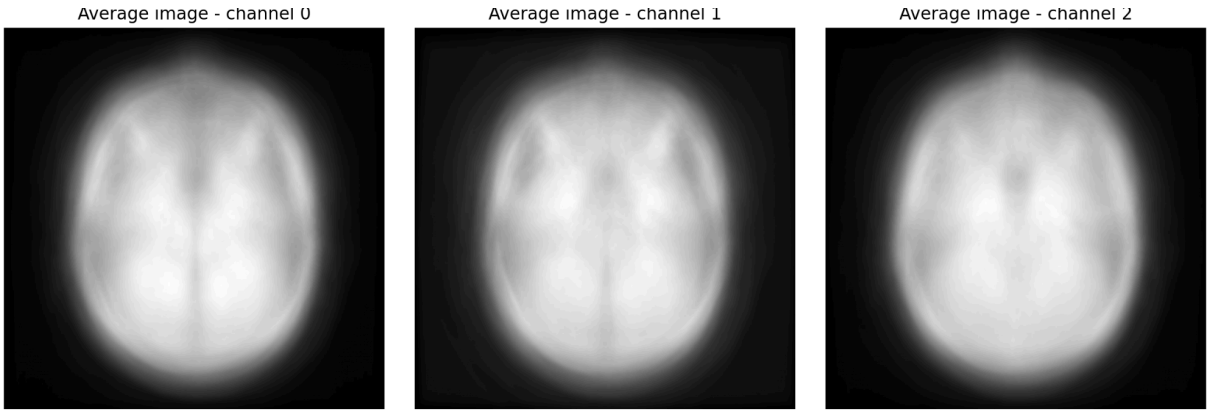
- **Race:** demographic category reported for the patient.
- **Ethnicity:** ethnic background reported for the patient.
- **Survival-related information:** variables associated with patient outcome, such as survival status or death indicator.

This metadata is not the direct target of the segmentation problem, but it is useful for understanding the patient and supporting complementary analyses.

2.6 Exploration

2.6.1 Average Image

The following image is a recreation of the average image of the dataset for the three channels.

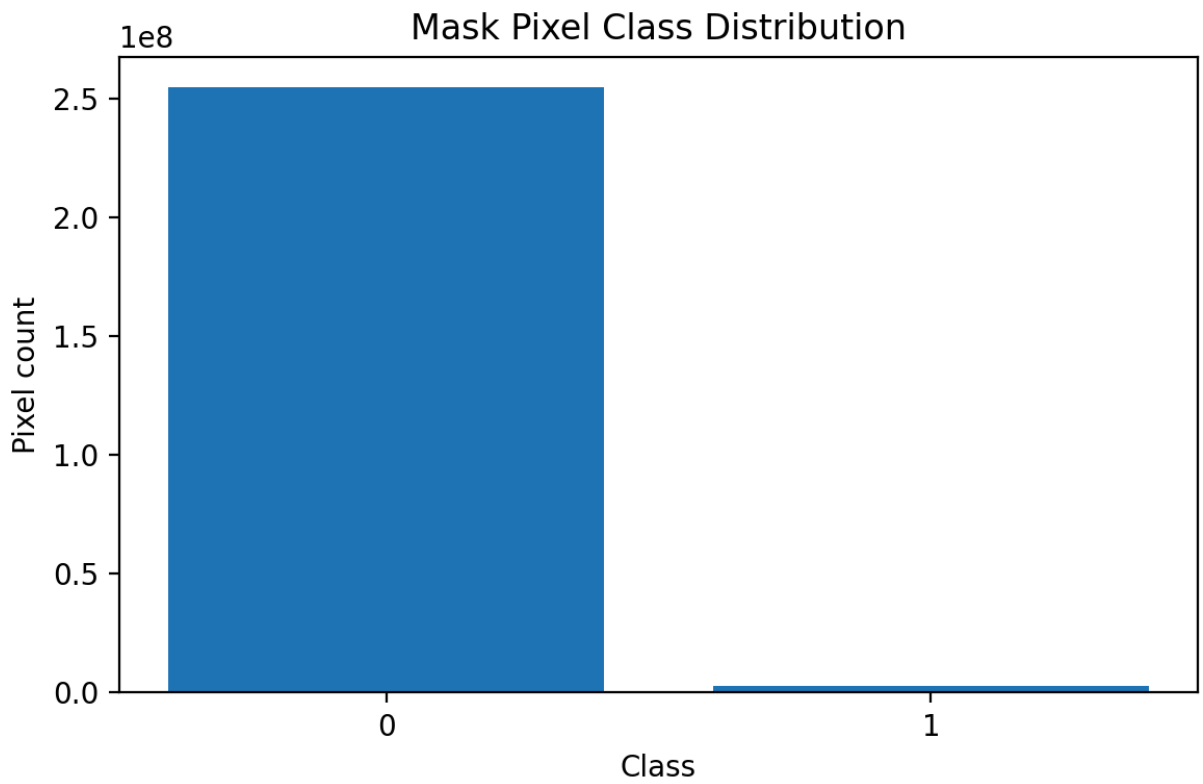


2.6.2 Class distribution

For the image segmentation task, the labels are provided through the segmentation masks, where each pixel belongs to one of two classes.

- **Class 0:** Represents the background, meaning pixels that do not correspond to the abnormal region and therefore are not the target of interest.
- **Class 1:** Represents the FLAIR abnormality region, which is the area that the model is expected to detect and segment.

Pixel class distribution		
Class	Number of pixels	Percentage
0	254,839,090	98.97%
1	2,651,854	1.03%



We can note that the dataset is far from being balanced, this due the nature of the dataset and difficult to get new instances.

2.7 Data quality

The quality of the dataset is appropriate for a medical image segmentation task, since it provides a consistent structure across samples and includes manually annotated masks that serve as reliable ground truth. All MRI slices have the same spatial resolution of 256 × 256 pixels, and the images are stored in a uniform 3-channel .tif format, which facilitates preprocessing and model training. In addition, each image has a corresponding segmentation mask, allowing a direct pairing between input data and target labels. However, the dataset also presents some challenges that must be considered during analysis, such as the strong class imbalance.

3.0 Data Preparation

3.1 Data split

To split the dataset, a patient-level strategy was selected in order to avoid data leakage. Since each patient contains multiple MRI slices, all images and masks from the same patient must remain in the same subset.

First, a final test set is reserved using approximately 10–15% of the 110 patients. The remaining patients are then used in a 5-fold cross-validation scheme. In each fold, 4 parts are used for training and 1 part is used for validation. This approach allows the model to be validated several times while keeping an independent final test set for the final evaluation.

Stage	Distribution	Approximate patients	Purpose
Final test set	10–15%	11–17	Final unbiased evaluation
Remaining data	85–90%	93–99	Used for cross-validation
Training in each fold	4/5 of remaining data	about 74–79	Model training
Validation in each fold	1/5 of remaining data	about 19–20	Model validation

3.2 Data augmentation

In order to improve the robustness of our data set, the next data augmentation modifications will be applied for the training data, with

Data augmentation	Description / purpose
Horizontal flip	Introduces spatial variation in the image orientation and helps the model generalize better.
Small rotations	Simulates slight changes in image orientation during acquisition while preserving the anatomical structure.
Small shifts / translations	Moves the image slightly in horizontal or vertical directions to improve robustness to position changes.
Small scaling / zoom	Helps the model adapt to small variations in the apparent size of the abnormal region.
Brightness adjustment	Simulates differences in image intensity between scans.
Contrast adjustment	Improves robustness to variations in tissue contrast and scanner conditions.

The selected augmentations are not intended to be applied aggressively. Instead, they are used with small magnitudes and controlled probability. In this way, augmentation increases variability in the training data without introducing unrealistic distortions.

3.3 Prepared dataset

After the data augmentation applied to training data and the separation into folds, the ramming processed data follows the next structure:

Stage / Fold	Subset	Patients	Image-mask pairs
Final split	Train + Validation pool	99	3599
	Final Test	11	330
Fold 1	Train	79	2809
	Validation	20	790
Fold 2	Train	79	2784
	Validation	20	815
Fold 3	Train	79	2926
	Validation	20	673
Fold 4	Train	79	2942
	Validation	20	657
Fold 5	Train	80	2935
	Validation	19	664

4.0 Modeling

The modeling phase was divided into N iterations, where each one has a different purpose in the project, since they have a baseline to compare the architecture proposals and the improvement of those depending on the results of each iteration. This approach is pretended to improve the model and get better results.

4.1 Threshold-based segmentation

A segmentation threshold was selected, a very basic baseline that is not pretended to get accurate results, and is a start point for comparison against the next models. Threshold segmentation is an appropriate

baseline for image segmentation because it is a simple, interpretable, and computationally inexpensive method that directly separates foreground from background based on pixel intensity values. The disadvantage of this approach is that it does not learn patterns or spatial context from data.

Note: Since this approach is not a trainable model, there is not weights saved, instead the configuration was saved as a JSON, the information can be consulted on the repostiry,

4.1.1 Parameters and configuration

Aspect	**Configuration**
Model name	threshold_baseline
Model type	Classical non-trainable baseline
Segmentation approach	Fixed threshold segmentation
Input image type	Multi-channel .tif image
Channel used for segmentation	Channel index 1
Channel name used	FLAIR
Normalization	Min-max normalization per image
Threshold value	0.55
Output	Binary mask
Output classes	0 = background, 1 = lesion
Training required	No
Learned weights	No
Cross-validation folds	5
Validation strategy	Fold-based validation on fold 1 to fold 5
Final evaluation	Separate final test split

4.2 Encoder - decoder V1

The architecture selected for a more solid model was an encoder decoder, this because it is specifically designed to solve the two main requirements of the task: understanding the content of the image and recovering the spatial location of each region at pixel level. The encoder progressively transforms the input image into a compact set of high-level feature representations, allowing the model to capture relevant patterns, textures, and contextual information. The decoder then upsamples these learned representations to recover the spatial resolution needed to generate a pixel-wise segmentation mask.

Despite the advantages, this architecture represents some limitations to consider such as the loss of spatial detail during downsampling, possible coarse boundary reconstruction, higher computational cost, sensitivity to hyperparameters, and risk of overfitting on limited datasets.

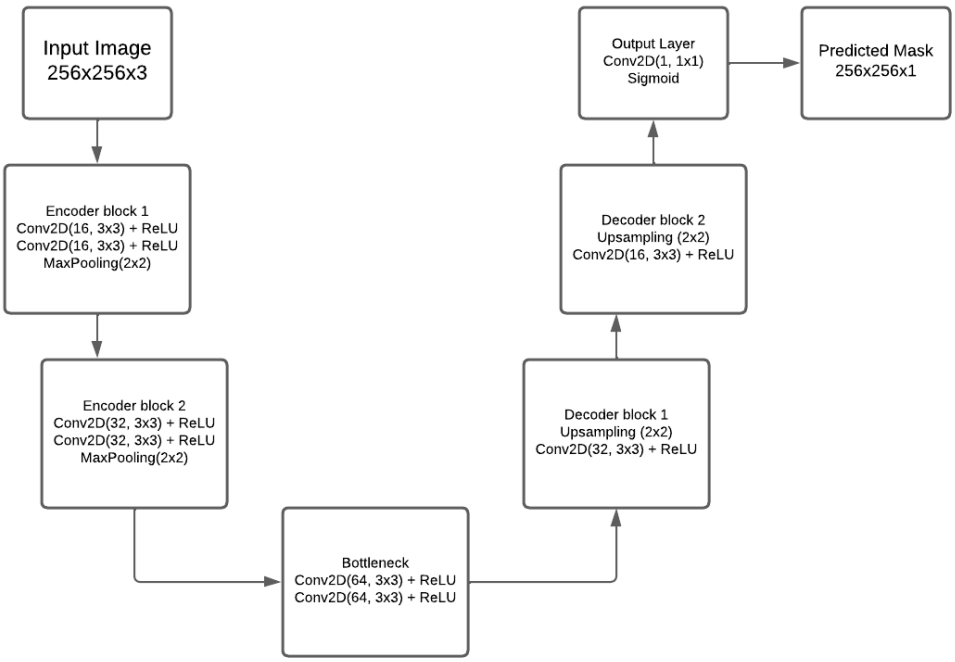
4.2.1 Parameters and configuration

Aspect	**Configuration**
Model name	encoder_decoder_v1
Framework	Keras / TensorFlow
Task	Binary image segmentation
Input shape	256 x 256 x 3
Output shape	256 x 256 x 1
Output activation	Sigmoid
Loss function	Binary Cross-Entropy
Optimizer	Adam
Learning rate	1e-3
Batch size	8
Epochs	15
Prediction threshold	0.5
Cross-validation folds	5
Random seed	42
Image normalization	Per-channel min-max normalization

Checkpoint strategy	Best model saved by val_loss

4.2.2 Architecture

This first version has a basic configuration with two main encoding and decoding blocks, using convolutional layers followed by ReLU activation functions to extract nonlinear features from the input image. In the encoder, max-pooling layers are applied to progressively reduce the spatial dimensions and capture higher-level contextual information. In the decoder, upsampling layers are used to restore the spatial resolution and reconstruct the segmentation mask. Finally, a 1x1 convolution with sigmoid activation produces a pixel-wise binary output, where each pixel is classified as background or lesion. This configuration represents a basic standard for modern encoder-decoder architectures and leaves room for possible improvements.



This architecture was directly inspired by the image segmentation encoder-decoder model described by Kayasth [5], as well as the U-Net architecture introduced by Ronneberger, Fischer, and Brox [8].

4.3 Encoder - decoder V2

There are two main objectives for this architecture based on the previous version results. The first one is to improve the model through getting a better score for IoU, precision and recall mainly against the version 1; the second one is to make it more computational efficient, this due the long time that takes training the previous architecture, the previous architecture took 6 hours to train. Further are explained all the changes and why that decision was taken.

4.3.1 Changes and improvements

Considering the objectives for this new version of the previous model, the changes applied to the parameters and architecture were strongly inspired by the U-Net architecture proposed by Ronneberger and Fischer. The main new element taken from U-Net is the implementation of skip connections, which allow feature maps from the encoder to be directly transferred to the decoder. This helps preserve spatial information that is usually lost during downsampling and improves the model's ability to recover fine details in the segmented region.

The changes are:

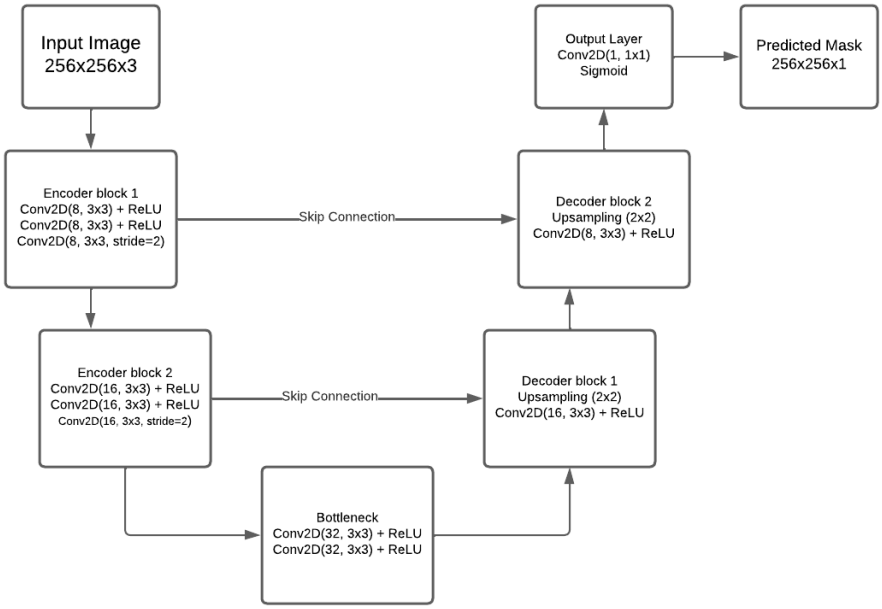
Change	Base model V1	New version (V2)	Why the change was made
Skip connections	No	Yes	To preserve spatial information from the encoder and pass it directly to the decoder, improving localization of the lesion region.
Downsampling method	Max Pooling	Strided convolution	To make downsampling learnable, so the network can learn how to reduce spatial resolution while extracting useful features. This was successfully my previous personal work [9], where adding as tried convolutions instead max pooling can improve the patterns learned. In this case it is thinked to compensate for the filter reduction.
Number of filters	16 → 32 → 64	8 → 16 → 32	To make the model lighter and reduce computational cost while keeping a hierarchical feature representation.
Decoder design	Upsampling + convolution	Upsampling + skip connection + convolution	To improve reconstruction of the output mask by combining decoder features with encoder features.
Encoder design	Convolution blocks + max pooling	Convolution blocks + skip connections + strided convolutions	To preserve more relevant spatial information and make downsampling learnable.
Batch size	8	16	To evaluate whether a larger batch size can improve training speed and

			computational efficiency if memory allows.
Loss function	Binary Cross Entropy	Dice Loss	To address class imbalance more effectively and optimize overlap between predicted and true lesion regions.

4.3.2 Parameters and configuration

Aspect	Configuration
Model name	encoder_decoder_v2
Framework	Keras / TensorFlow
Task	Binary image segmentation
Input shape	256 x 256 x 3
Output shape	256 x 256 x 1
Output activation	Sigmoid
Loss function	Dice Loss
Optimizer	Adam
Learning rate	1e-3
Batch size	16
Epochs	15
Prediction threshold	0.5
Cross-validation folds	5
Random seed	42
Image normalization	Per-channel min-max normalization
Checkpoint strategy	Best model saved by val_dice_coefficient

4.3.3 Architecture



4.4 Encoder - decoder V3

Due to the failure of the previous version, this new one keeps the same U-Net like architecture as the previous one but with some changes to the architecture and parameters. The objective of this version is not to have a more cost-efficient model, this could be addressed in future models; The objective gets a real improvement against the metrics of the version 1.0.

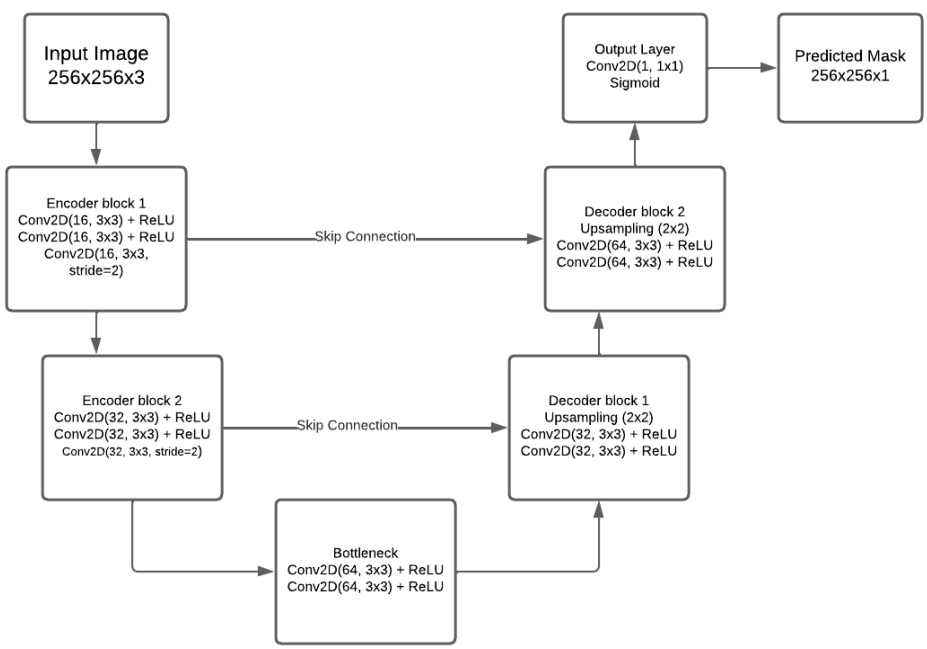
4.4.1 Changes and improvements

Change	Previous version (V2)	New version (V3)	Why the change was made
Number of filters	8 → 16 → 32	16 → 32 → 64	To recover representational capacity, since the reduced filters in V2 likely made the model too weak to learn foreground lesion patterns robustly.
Decoder capacity	16 → 8	32 → 16	To make the decoder consistent with the stronger encoder and improve reconstruction of lesion regions in the output mask.

Decoder convolution depth	One convolution after each skip concatenation	Two convolutions after each skip concatenation	To strengthen feature refinement in the decoder after merging encoder and decoder information.
Loss function	Dice loss	BCE + Dice loss	To improve optimization stability, since pure Dice loss may favor unstable training or collapse in highly imbalanced segmentation settings.
Batch size	16	8	To make updates less coarse and closer to the baseline setting, potentially improving learning stability.
Prediction threshold strategy	0.5 only	0.5 as default + sweep of 0.5, 0.3, 0.25, 0.2	To evaluate whether the model predicts useful foreground probabilities below 0.5 and select a better operating threshold.

4.4.2 Parameters and configuration

4.4.3 Architecture



4.5 nnU-Net Architecture

As a final point of comparison, it was decided to implement this nnU-Net architecture (this is not the exact framework, is a reimplementation inspired from the original architecture) a considered state of art architecture by F. Isensee, J. Petersen, A. Klein, D. Zimmerer, P. F. Jaeger, S. Kohl, J. Wasserthal, G. Kohler, T. Norajitra, S. W., and K. H. Maier-Hein from the Division of Medical Image Computing, German Cancer Research Center (DKFZ) [10]. This architecture was designed specifically for medical image segmentation so it attends the main problem of this project. The official project says it works particularly well in training-from-scratch settings, including biomedical and other non-standard imaging problems, where off-the-shelf natural-image pretrained models are often not a great fit. That matters a lot for tasks like your MRI lesion segmentation.

Note: Due the computational cost of this architecture, this was trained on Google Colab using the acceleration GPU and the dataset split was changed to a traditional s

4.5.1 Parameters and configuration

4.5.1 Architecture

5.0 Evaluation

The evaluation section aims to measure the segmentation performance of each model in a consistent and comparable way. Since all approaches in this study produce a predicted segmentation mask, the same evaluation metrics can be applied across classical and deep learning models. To assess performance, the metrics selected were Intersection over Union (IoU), Precision, Recall, and Pixel Accuracy.

- **IoU:** The primary metric because it directly measures the agreement between predicted and ground-truth masks.
- **Recall:** Also highly relevant, especially in lesion segmentation, where missing true lesion pixels is undesirable.
- **Precision:** Helps identify over-segmentation.
- **Pixel Accuracy:** Reported only as a complementary metric due to the strong class imbalance.

5.1 Threshold-based segmentation

5.1.1 Results

Fold / Split	TP	FP	TN	FN	IoU	Precision	Recall	Pixel Accuracy
Fold 1	122,657	1,611,309	49,589,050	450,424	0.0562	0.0707	0.2140	0.9602
Fold 2	88,692	1,816,349	51,072,561	434,238	0.0379	0.0466	0.1696	0.9579

Fold 3	53,812	1,442,680	42,344,552	264,684	0.0306	0.0360	0.1690	0.9613
Fold 4	127,653	1,705,901	40,820,137	403,461	0.0571	0.0696	0.2404	0.9510
Fold 5	127,238	1,872,837	41,166,491	349,338	0.0542	0.0636	0.2670	0.9489
Global CV	520,052	8,449,076	224,992,791	1,902,145	0.0478	0.0580	0.2147	0.9561
Final Test	61,434	681,908	20,715,315	168,223	0.0674	0.0826	0.2675	0.9607

5.1.2 Metrics analysis

- **IoU:** This is the most important metric in this baseline analysis because it directly measures overlap between the predicted mask and the ground truth. The obtained values are very low, which indicates poor segmentation quality.

- **Precision:** Precision is also very low, showing that a large proportion of the pixels predicted as lesions actually correspond to background. This suggests many false positives.

- **Recall:** Recall is slightly higher than Precision, meaning that the method is able to detect part of the lesion area, but still misses a large amount of true lesion pixels.

- **Pixel Accuracy:** Although Pixel Accuracy is high, it must be interpreted carefully because the dataset contains a large majority of background pixels. For that reason, this metric is less informative than IoU for this task.

5.1.3 Folds and test analysis

Overall, the behavior across folds suggests that thresholding is unstable and limited for this problem, since performance changes between folds but remains weak in all cases.

The final test slightly improved over global cross-validation, but performance remained low overall. This confirms that thresholding can serve as a basic reference baseline, yet it is not strong enough for accurate lesion segmentation due to its limited overlap and low precision.

5.1.4 Conclusions

These results confirm that threshold segmentation is useful as a simple baseline, but its low overlap and low precision demonstrate that more advanced learning-based models are necessary for this problem.

5.2 Encoder - decoder V1

5.2.1 Results

Fold / Split	TP	FP	TN	FN	IoU	Precision	Recall	Pixel Accuracy
Fold 1	255,563	129,210	51,071,149	317,518	0.3639	0.6642	0.4459	0.9914
Fold 2	313,217	75,903	52,813,007	209,713	0.5230	0.8049	0.5990	0.9947
Fold 3	218,421	79,043	43,708,189	100,075	0.5494	0.7343	0.6858	0.9959
Fold 4	357,662	103,124	42,422,914	173,452	0.5639	0.7762	0.6734	0.9936
Fold 5	225,841	22,261	43,017,067	250,735	0.4527	0.9103	0.4739	0.9937
Global CV	1,370,704	409,541	233,032,326	1,051,493	0.4841	0.7700	0.5659	0.9938
Final Test	175,608	80,671	21,316,552	54,049	0.5659	0.6852	0.7647	0.9938

5.2.2 Metrics analysis

- **IoU:** IoU shows a clear improvement compared with the threshold baseline, indicating that the encoder-decoder achieves a much stronger overlap between predicted masks and ground truth.

- **Precision:** Precision is high in most folds, which means that a large proportion of the predicted lesion pixels are correct.

- **Recall:** Recall is also considerably higher than in the threshold baseline, showing that the model is able to recover a larger portion of the true lesion area.

- **Pixel Accuracy:** Pixel Accuracy remains very high, but as in the previous case, it should be interpreted carefully because background pixels dominate the images.

5.2.3 Folds and test analysis

The encoder-decoder shows substantially better and more consistent performance than the threshold baseline across all folds. Folds 3 and 4 achieved the strongest balance between Precision and Recall, resulting in the highest IoU values. Overall, the fold results show that the encoder-decoder is capable of learning meaningful lesion patterns, although some variability between folds is still present.

The combination of a higher Recall and solid IoU in test suggests that the model produced a more complete lesion segmentation on the final test set. This confirms that the encoder-decoder provides a much more effective segmentation approach than the threshold baseline and establishes a solid first deep learning benchmark.

5.2.4 Conclusions

These results show that the encoder-decoder significantly outperforms the threshold baseline and provides a strong initial deep learning solution for lesion segmentation. However this model can get a better performance and according to the previous results the work to be done in order to increase the performance of our model but not significantly considering the computational cost.

5.3 Encoder - decoder V2

5.3.1 Results

Training was stopped because the model showed clear evidence of early collapse to background prediction. After the first epoch, both training and validation metrics became completely flat, which indicates that the optimization process was no longer producing meaningful learning. In addition, the per-image validation results showed TP = 0 and FP = 0 across the reported samples, meaning that the model predicted only background pixels. For slices with actual lesion regions, this caused FN > 0, IoU = 0, Precision = 0, and Recall = 0, which confirms that the model was not segmenting the foreground class.

Aspect	Observed result	Interpretation
Training Dice	0.4896 → 0.6576 at epoch 1, then flat until epoch 14	The model improved only at the beginning and then stopped changing
Training Loss	0.5106 → 0.3425 at epoch 1, then flat until epoch 14	Optimization stalled very early
Validation Dice	0.6350 from epoch 0 to epoch 14	No validation improvement at all
Validation Loss	0.3696 from epoch 0 to epoch 14	The model remained in the same solution during the whole training
True Positives (TP) on shown validation images	0 in all reported cases	The model did not detect foreground pixels in those samples
False Positives (FP) on shown validation images	0 in all reported cases	The model predicted no lesion pixels at all

True Negatives (TN)	Very high, often 65536	The model mostly classified everything as background
False Negatives (FN)	Greater than 0 in positive-mask slices	Real lesion pixels were missed completely
IoU	0.0 in all shown rows	No overlap between prediction and true lesion in positive cases
Precision	0.0 in all shown rows	No positive predictions were produced
Recall	0.0 in all shown rows	The model failed to recover lesion pixels
Pixel Accuracy	Very high, around 0.9888 to 1.0000	Misleadingly high due to class imbalance and dominant background
Training time	3.75 hours	That was expected due the filter reductions but it has consequences with the rest of metrics and how learned the model.

5.3.4 Conclusions

Based on this behavior, continuing the same training was not justified, since the model had already converged to a poor trivial solution and was not improving across epochs.

Considering this, the main objective is back to a configuration that gets better performance sacrificing or leaving for further models or versions the optimization of the model. A proposed decisions is:

Improvement	**Reason**
Restore filter sizes closer to V1	To recover model capacity and improve feature extraction
Keep skip connections	To preserve spatial information and support localization
Test lower thresholds such as 0.4 or 0.3	To avoid discarding weak but useful foreground probabilities
Consider BCE + Dice loss	To improve optimization stability and reduce collapse to all-background predictions

5.4 Encoder - decoder V3

5.4.1 Results

fold	TP	FP	TN	FN	IoU	Precision	Recall	Pixel Accuracy
fold_1	309828	716274	50484085	263253	0.240297	0.301947	0.540636	0.981081
fold_2	126591	43692	52845218	396339	0.223413	0.743415	0.242080	0.991762
fold_3	109565	174853	43612379	208931	0.222084	0.385225	0.344007	0.991299
fold_4	176831	50009	42476029	354283	0.304292	0.779541	0.332944	0.990610
fold_5	222131	81333	42957995	254445	0.398149	0.731985	0.466098	0.992284
global CV	944946	1066161	232375706	1477251	0.270886	0.469864	0.390119	0.989217
fold_5	38053	53730	17975880	151345	0.156514	0.414597	0.200916	0.988744

5.4.2 Metrics analysis

- **IoU:** The values are considerably lower than those obtained with the encoder-decoder V1. Although the model is able to detect lesion regions, the overlap between predicted masks and ground truth remains limited in most folds. The global cross-validation IoU of 0.270886 and the final test IoU of 0.156514 indicate that this version did not achieve a strong segmentation quality.

- **Precision:** Precision is highly variable across folds. In folds 2, 4, and 5, the model achieved relatively high Precision, which means that when it predicts lesion pixels, many of them are correct. However, fold 1 shows a very low Precision due to a high number of false positives, indicating probably that the model tends to over-segment in some cases.

- **Recall:** Recall is generally low to moderate. Fold 1 obtained the highest Recall, but this came together with a very large number of false positives. In the other folds, Recall remained limited, showing that the model is still missing a substantial portion of the true lesion area. This issue becomes more evident in the final test, where Recall drops to 0.200916, suggesting that the model fails to recover much of the lesion region on unseen data.

- **Pixel Accuracy:** As in the previous models, this metric must be interpreted carefully because background pixels dominate the images. Therefore, a high Pixel Accuracy does not necessarily imply a good lesion segmentation performance.

5.4.3 Fold and test analysis

The results show that this version presents a high variability across folds and does not maintain a stable balance between Precision and Recall. Fold 1 obtained the highest Recall, but its very low Precision indicates strong over-segmentation. In contrast, folds 2 and 4 reached high Precision values, but their Recall remained low, meaning that the model produced conservative predictions and missed many lesion pixels. Fold 5 achieved the best overall balance among the validation folds, resulting in the highest IoU value of 0.398149, although it is still below the values reached by the previous version 1.

5.4.4 Conclusions

The results did not achieve the expected results and the goal, against all the waited adjustments for a better performance, taking a strong inspiration from a U-Net architecture, the results given are worse than the most basic version and the time that takes to train this model was about ten hours, four more than the version 1.

Although the results are worse, this is not necessarily something bad, this model could be more conservative and we need to consider the unbalance of classes, and this model could lead to more true positives and less false positives.

6.0 Predictions

7.0 General Conclusions

References

[1] M. Buda, “Brain MRI segmentation,” *Kaggle*, [Online]. Available: https://www.kaggle.com/datasets/mateuszbuda/lgg-mri-segmentation/data. [Accessed: Apr. 12, 2026].

[2] K. Alomar, H. I. Aysel, and X. Cai, “Data augmentation in classification and segmentation: A survey and new strategies,” *Journal of Imaging*, vol. 9, no. 2, p. 46, 2023. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC9966095/. [Accessed: Apr. 12, 2026].

[3] GeeksforGeeks, “Thresholding-Based Image Segmentation,” *GeeksforGeeks*. [Online]. Available: https://www.geeksforgeeks.org/software-engineering/thresholding-based-image-segmentation/. [Accessed: Apr. 15, 2026].

[4] GeeksforGeeks, “SegNet - A Deep Convolutional Encoder-Decoder Architecture for Image Segmentation,” *GeeksforGeeks*. [Online]. Available: https://www.geeksforgeeks.org/computer-vision/segnet-a-deep-convolutional-encoder-decoder-architecture-for-image-segmentation/. [Accessed: Apr. 17, 2026].

[5] J. Kayasth, “Image Segmentation using Encoder-Decoder model,” preprint, Dec. 2022. [Online]. Available: ResearchGate. [Accessed: Apr. 20, 2026.]

[6] V. Badrinarayanan, A. Kendall, and R. Cipolla, “SegNet: A deep convolutional encoder-decoder architecture for image segmentation,” *arXiv preprint* arXiv:1511.00561, Oct. 2016.

[7] G. Xu, X. Wang, X. Wu, X. Leng, and Y. Xu, “Development of skip connection in deep neural networks for computer vision and medical image analysis: A survey,” *arXiv preprint* arXiv:2405.01725, May 2024. [Accessed: Apr. 20, 2026.]

[8] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: Convolutional Networks for Biomedical Image Segmentation,” *arXiv preprint* arXiv:1505.04597, May 2015. [Accessed: Apr. 20, 2026.]

[9] J. Ramirez, “Multi-Label Image Classification of Pokemon Types Using Deep Learning” in *Reto-ML*, GitHub repository. [Online]. Available: https://github.com/j1rd2/Reto-ML/blob/main/report.pdf. Accessed: Apr. 20, 2026. [Accessed: Apr. 20, 2026.].